Depressive Disorder Analysis using Gamma EEG Data

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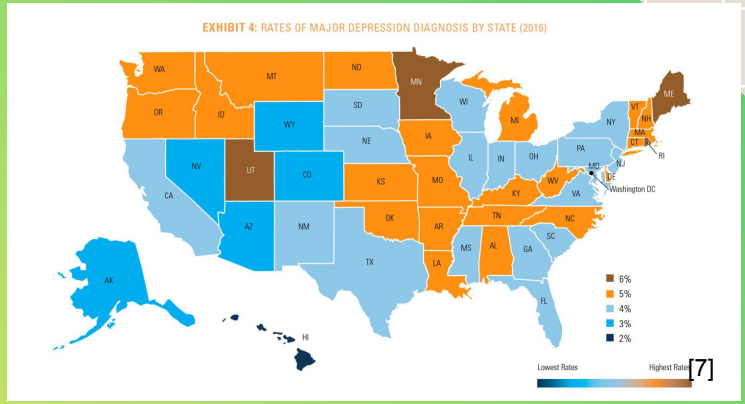
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Background

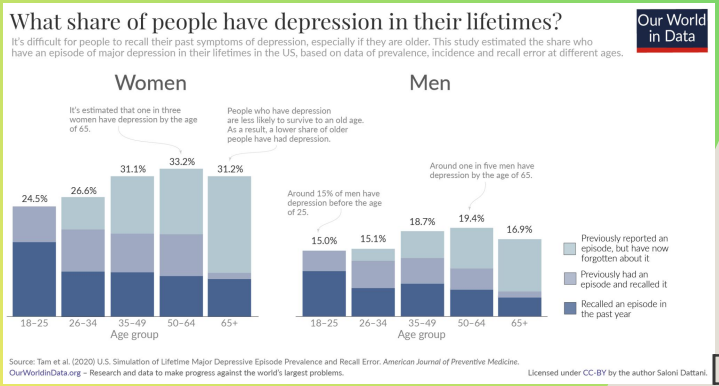
- 5.7% of adults suffer from depressive disorder globally [1]
- Symptoms [1]
  - Loss of interest in activities previously enjoyed
  - Depressed mood
  - Poor concentration
  - Suicidal thoughts
- Depressive disorder can affect relationships, work, school[1]
- Treatment options:
  - Behavioural therapy
  - Interprersonal psychotherapy
  - Antidepressant medications (SSRIs) [2]
- Important to diagnose the disorder and have personalized treatment options as this can reduce suicide risk by over 60% [3]

Background

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[7]



[8]

Background

Current methods of diagnosis include:

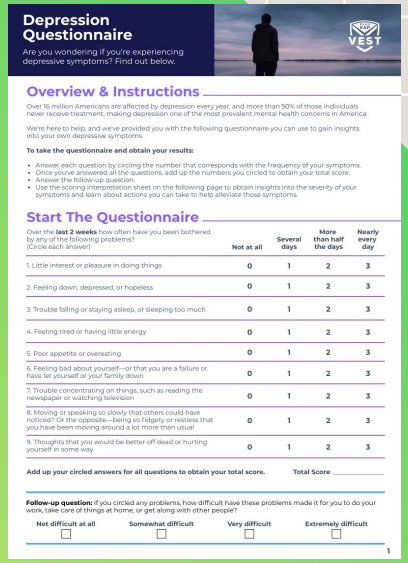
- Physical examination [4]
- Lab tests [5]
- Pyschiatric evaluation [6]

Limitations: [6]

- Self reporting symptoms
- Symptoms can have comorbidity with other psychiatric disorders
- Individuals can have differing symptoms

Our Proposed Method

- EEG detects depression non-invasively without requiring self reported symptoms
- The model is trained to resting state EEG data so the patient does not need to do any extra tasks
- The method provides unbiased evaluation of brain activity



Depression Questionnaire

Are you wondering if you're experiencing depressive symptoms? Find out below.

VEST

Overview & Instructions

Over 16 million Americans are affected by depression every year, and more than 50% of those individuals never receive treatment, making depression one of the most prevalent mental health concerns in America.

We're here to help, and we've provided you with the following questionnaire you can use to gain insights into your own depressive symptoms.

To take the questionnaire and obtain your results:

- Answer each question by circling the number that corresponds with the frequency of your symptoms.
- Once you've answered all the questions, add up the numbers you circled to obtain your total score.
- Answer the follow-up question.
- Use the scoring interpretation sheet on the following page to obtain insights into the severity of your symptoms and learn about actions you can take to help alleviate those symptoms.

Start The Questionnaire

Over the **last 2 weeks** how often have you been bothered by any of the following problems? (Circle each answer)

	Not at all	Several days	More than half the days	Nearly every day
1. Little interest or pleasure in doing things	0	1	2	3
2. Feeling down, depressed, or hopeless	0	1	2	3
3. Trouble falling or staying asleep, or sleeping too much	0	1	2	3
4. Feeling tired or having little energy	0	1	2	3
5. Poor appetite or overeating	0	1	2	3
6. Feeling bad about yourself—or that you are a failure or have let yourself or your family down	0	1	2	3
7. Trouble concentrating on things, such as reading the newspaper or watching television	0	1	2	3
8. Moving or speaking so slowly that others could have noticed? Or the opposite—being so fidgety or restless that you have been moving around a lot more than usual	0	1	2	3
9. Thoughts that you would be better off dead or hurting yourself in some way	0	1	2	3

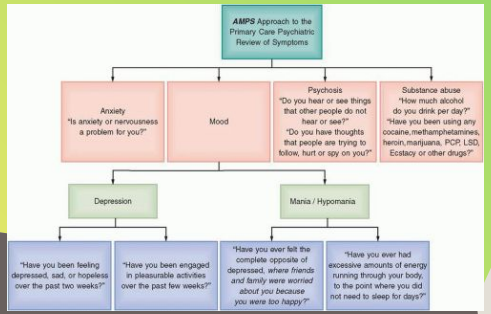
Add up your circled answers for all questions to obtain your total score. **Total Score** _____

Follow-up question: If you circled any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?

Not difficult at all ☐ Somewhat difficult ☐ Very difficult ☐ Extremely difficult ☐

1

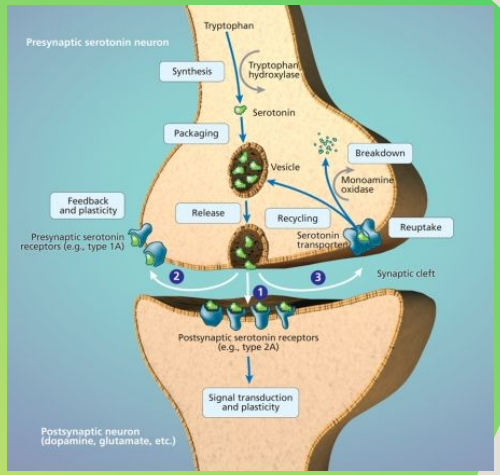
[9]



[10]

Neurophysiological Ideas

- Serotonin synapse
  - An enzyme tryptophan hydroxylase
- The serotonin transporter gene variation
  - 1 short and 1 long allele
- Stress
  - Short alleles is vulnerable to depression under stress
- DMN Hypo-Connectivity
  - Decrease in connectivity between the cortexes + lobules have been shown to be associated with MDD.



[11]

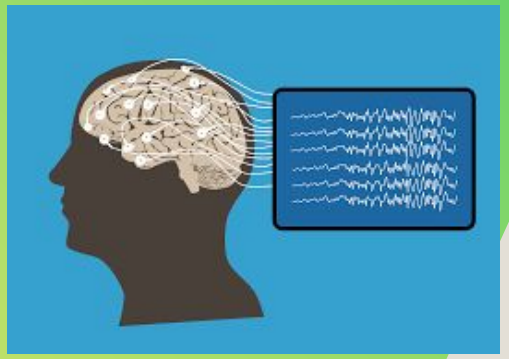
Using EEG Data for Depressive Disorder

<u>Previous EEG biomarkers investigated</u>

- Previous studies proposed that frontal alpha asymmetry was a biomarker, but was proven inconsistent
- P300 Amplitude reductions

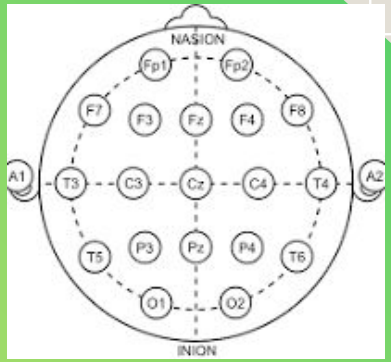
<u>Emerging Biomarkers</u>

- Reduced gamma activity was consistent
- The reduced activity was consistent even with varying symptom severity, medication, comorbid anxiety, and demographic factors
- Emerging evidence suggests reduced gamma power may be influenced by oxidative stress and neuroinflammatory processes
- It is thought that gamma activity decreases may present the electrophysiological symptoms of DMN



[12]

Data Set Background [13], [15]



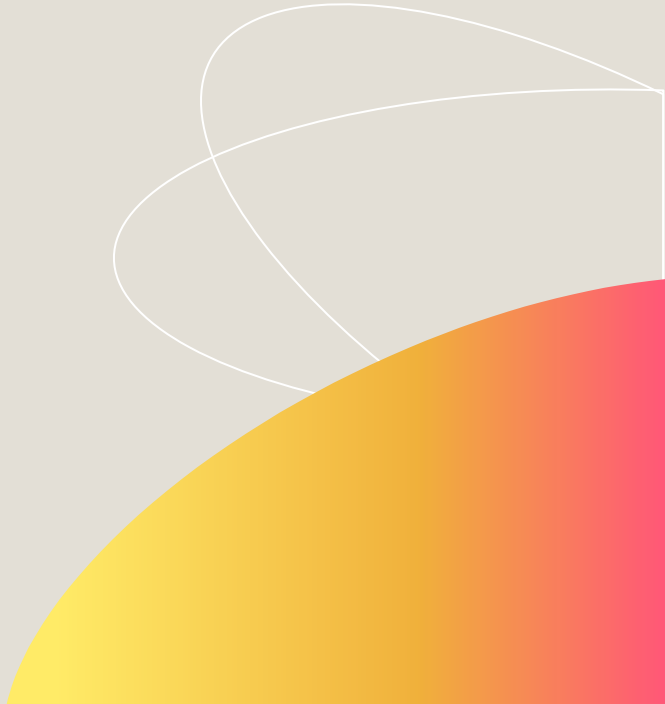
- EEG taken at resting state with eyes closed for 5 minutes. Electrode setup is the 10-20 montage.
- Digital EEG data filtered using a butterworth high pass filter
- Converted in to frequency domain using FFT with the following parameters:
  - Epoch = 2s
  - Sample rate = 128 samples/s
  - Frequency range = 0.5-40 Hz
  - Resolution 0.5 Hz
  - Cosine taper window to minimize leakage
- Coherence and power spectral density extracted from all bipolar channels

$$coherence\ (f) = \frac{\left(\sum_N (a(x)u(y) + b(x)v(y))\right)^2 + \left(\sum_N (a(x)v(y) + b(x)u(y))\right)^2}{\sum_N (a(x)^2 + b(x)^2) \sum_N (u(y)^2 + v(y)^2)}$$

[13]

Goal of Project

- Create a machine learning algorithm to diagnose depressive disorder using EEG data at resting state with gamma power bands



Our Data Processing Steps



Algorithm Development Process

1. Convolutional and Graphical Neural Networks

2. Convolutional Neural Network with XG Boost

3. Multilayer Percption (MLP)

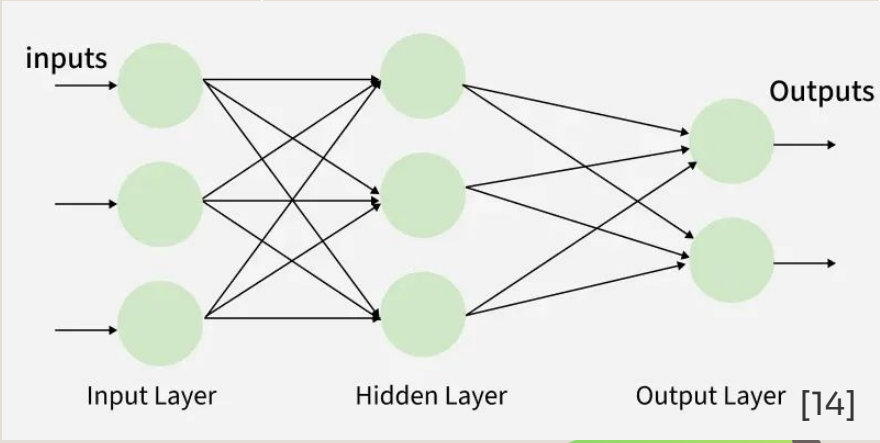
4. Random Forest Model

5. XGBoost Model

6. Feature down select and regularization

Multilayer Perceptron (MLP)

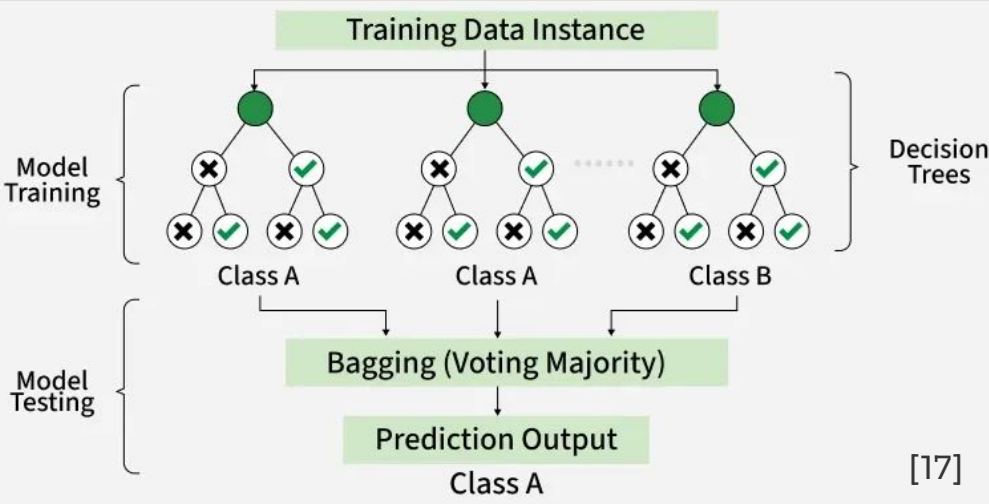
- Feed forward neural network
- Input
  - Gamma power
  - Gamma coherence
- Output
  - Depressed patient (1)
  - Healthy control (0)
- Removed participant number, age and IQ
- Struggled with our dataset
  - Receiver Operating Characteristic - Area Under Curve (ROC - AUC): 49%
  - 0 recall: 32%
  - 1 recall: 74%



[14]

Random Forest Model

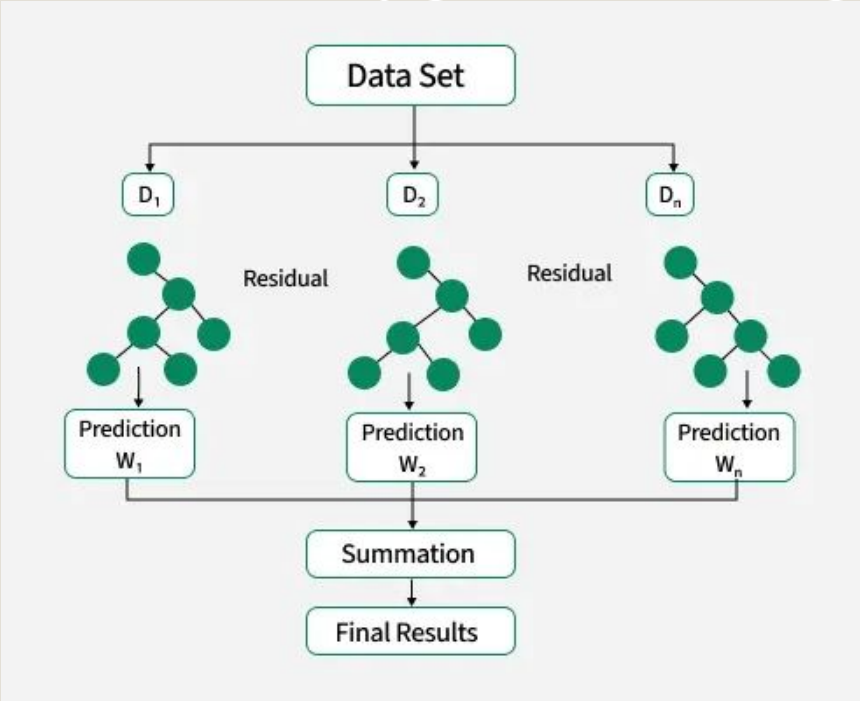
- Decision trees
- Input
  - Gamma power
  - Gamma coherence
- Output
  - Depressed patient (1)
  - Healthy control (0)
- Removed participant number, age and IQ
- Better performance
  - ROC-AUC: 72%
  - 0 recall: 44%
  - 1 recall: 76%
- Coherence was the most important feature



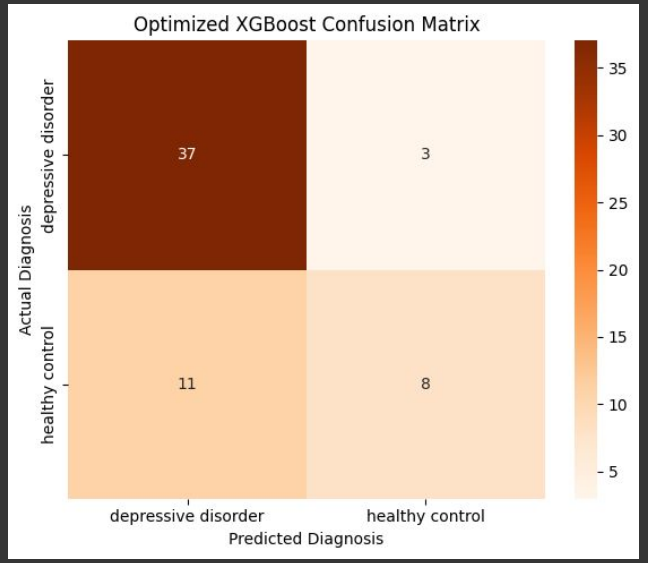
[17]

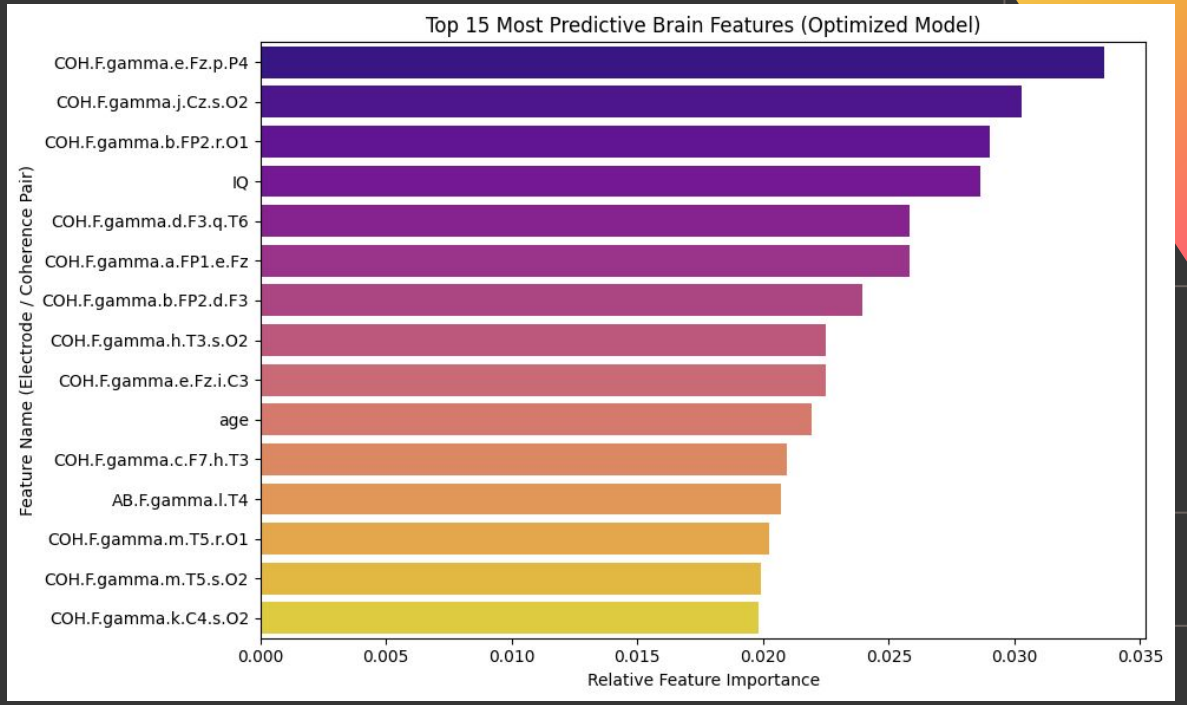
XGBoost Model

- Sequential Decision Trees
- Input
  - Gamma power
  - Gamma coherence
- Output
  - Depressed patient (0)
  - Healthy control (1)
- Better performance
  - ROC-AUC: 74%
  - 0 recall: 93%
  - 1 recall: 42%
- This model included the demographic data (IQ and age)



XGBoost Results





Comparison of Models

	ROC-AUC	Recall	d’
MLP	49%	Depressed: 74% Healthy: 32%	0.16
RF	72%	Depressed: 76% Healthy: 44%	0.769
XGB	74%	Depressed: 93% Healthy: 42%	1.240

Conclusion

- RF vs MLP vs XGB
  - Best ROC-AUC performance
  - All models were biased towards depressed patients
- IQ and age were highlighted as confounding factors
- Coherence > Power
- Parietal and Occipital Region Significance



Future Directions

- Increase the sample of healthy controls
- Introduce other frequency bands (such as delta, theta, alpha, etc.)
- Possibly explore other neural networks
- Larger sample size for better generalization
- Test models on new samples from a different population

Safety Considerations

- Inaccurate diagnosis of mental illnesses like depression can subject the patients to improper treatments

- EEG recordings are done on the scalp in a 10-20 montage configuration



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Any questions?
Ask away!


Thank you